

COURSE TITLE : INSTRUMENTATION IN PROCESS INDUSTRIES
COURSE CODE : 6215
COURSE CATEGORY : E
PERIODS/WEEK : 5
PERIODS/SEMESTER : 75
CREDITS : 5

TIME SCHEDULE

Module	Topics	Periods
1	Measurement And control in Paper industry	18
2	Measurement And control in Distillation column	19
3	Measurement And control in Petrochemical industry	19
4	Process And measurement in Cement industry	19
Total		75

COURSE OUTCOME

Module	G.O	On completion of the study of this Module the student will be able
1	1	To understand the measurement and control of parameters in Pulp manufacturing process
2	1	To understand the measurement and control of parameters in Distillation column
3	1	To understand the process in petrochemical industry
	2	To understand the instrumentation in petrochemical industry
4	1	Understand cement manufacturing process

MODULE I INSTRUMENTATION IN PAPER INDUSTRY

1.1.0 To understand the basic Pulp manufacturing process

- 1.1.1 To list the Raw Materials used
- 1.1.2 To explain the paper manufacturing process
- 1.1.3 To illustrate the measurement Hard ware in the paper Industry
- 1.1.4 To list the Analyzers
- 1.1.4 To explain the consistency analyzer using strain gauges
- 1.1.5 To explain Controllers and Displays in paper Industry
- 1.1.5 To describe Typical Control systems in paper industry.

MODULE II CONTROL OF DISTILLATION TOWER

2.1.0 To understand Distillation column

- 2.1.1 To describe the distillation process
- 2.1.2 List the variables in distillation operation
- 2.1.3 To describe the pressure control in distillation columns
- 2.1.4 To describe column feed temperature control
- 2.1.5 To explain the Reboiler control
- 2.1.6 To explain the application of chromatograph in distillation
- 2.1.7 To describe the Reflux control scheme in distillation column

MODULE III INSTRUMENTATION IN PETROCHEMICAL INDUSTRY

3.1.0 To understand the process in petrochemical industry

- 3.1.1 To explain petroleum exploration
- 3.1.2 To explain petroleum production and refining
- 3.1.3 To explain sub process of petroleum refining
- 3.1.4 To explain final product and byproduct of petroleum refining
- 3.1.5 To describe the constituents of crude oil

3.2.0 To understand the instrumentation in petrochemical industry

- 3.2.1 P&I Diagram of petroleum refinery
- 3.2.2 Control of distillation column in petrochemical industry

MODULE IV Measurement and control in Cement Manufacturing Industries

4.1.0 Understand cement manufacturing process

- 4.1.1 To describe the cement manufacturing process
- 4.1.2 Know the various raw materials used for cement manufacturing
- 4.1.3 To describe Material Handling equipments used in cement Industries .
- 4.1.4 To describe the working of Raw mill
- 4.1.5 To describe the Weigh feeders and weight measurement techniques
- 4.1.6 To explain the operation of kiln and kiln temperature measurement.
- 4.1.7 To explain the working of electro static precipitator.

CONTENT DETAILS

MODULE I

Pulp manufacturing process-Raw Materials used- description of the process-Measurement Hardware in the paper Industry-Analyzers- consistency analyzer using strain gauges .Controllers and Displays in paper Industry –Typical Control systems.

MODULE II

Distillation column-condenser-reboiler-inter heater-feed preheater-the distillation process-controlled variables in distillation operation –manipulated variables in distillation operation- pressure control in

distillation columns- column feed temperature control--Reboiler control - chromatograph in distillation -the Reflux control scheme in distillation column

MODULE III

Process in petrochemical industry--petroleum exploration- petroleum production and refining
-sub process of petroleum refining- final product and byproduct of petroleum refining- the constituents of crude oil- the instrumentation in petrochemical industry- P&I Diagram of petroleum refinery- control of distillation column

MODULE IV

Cement manufacturing process--Various raw materials used for cement manufacturing- Material Handling equipments used in cement Industries . the working of Raw mill- the Weigh feeders and weight measurement techniques - operation of kiln and kiln temperature measurement.- working of electro static precipitator.

REFERENCES

1. Dr Ram Prasad " Petroleum Refining Technology"Khanna Publisher,1 st Edition ,2000
2. Liptak B G " Instrumentation in process industries" Chilton Book Company ,1973
3. k krishnaswamy 2nd edition " Process control " New age international (p) Ltd
4. D patrabis " Principles of Process Control" 2nd editionTMH publishing company pvt.Ltd